Newly Born

AO-02 | Adult Obstetrics

Quick flow

01 Birth [EMT / AEMT / Paramedic]

Term? • breathing/crying? • good muscle tone?

02 All Yes [EMT / AEMT / Paramedic]

Warm, dry, stimulate, skin-to-skin, maintain temperature, and reassess.

03 Any No [EMT / AEMT / Paramedic]

Warm, dry, stimulate, position airway, and suction only if needed.

04 Apnea / Gasping / HR <100? [EMT / AEMT / Paramedic]

Begin BVM ventilation at 40-60/min and assess chest movement and heart-rate response.

05 HR Remains <60? [EMT / AEMT / Paramedic]

After effective ventilation, begin two-thumb encircling 3:1 compressions, use 100% oxygen, and escalate care.

06 Reassess + Transport [EMT / AEMT / Paramedic]

Maintain warmth, right-hand SpO_2, cardiac monitoring, glucose, and early notification.

Overview

- Most newly born infants require only warmth, drying, stimulation, and appropriate cord management.
- Effective positive-pressure ventilation is the most important intervention when respirations are inadequate or heart rate remains below 100/min.

Provider-Level Actions

Perform immediately indicated care at the highest authorized level present while protecting both maternal and neonatal physiology.

EMT

- Warm, dry, and stimulate; position the airway and use bulb suction only when secretions obstruct breathing.
- If apnea/gasping or heart rate below 100 persists, provide BVM ventilation at 40-60 breaths/min and correct mask seal, airway position, and technique until chest movement is present.
- If heart rate remains below 60 after 30-60 seconds of effective ventilation, begin two-thumb encircling chest compressions coordinated at a 3:1 ratio.
- Apply pulse oximetry to the right hand, wrist, or upper arm and prevent hypothermia throughout care.

AEMT

- Perform all EMT care plus authorized supraglottic airway support, IV/IO access, fluid and glucose treatment within current neonatal standing orders. Epinephrine while a pulse is present is Paramedic-only; AEMT epinephrine remains available for confirmed pulseless arrest within the approved arrest orders.
- Use cardiac monitoring to determine heart rate when available without interrupting ventilation or compressions.

Paramedic

- Perform all prior care plus advanced airway management when ventilation remains ineffective and protocol-directed neonatal resuscitation medications. Paramedic: give the specified epinephrine dose for persistent HR below 60 with a pulse despite effective ventilation and coordinated compressions.

- Evaluate persistent failure to respond for hypovolemia, pneumothorax, hypoglycemia, congenital disease, infection, or maternal medication effect.

Indications

- Infant in the immediate newborn period following field, home, or facility delivery.
- Newly born infant with apnea, gasping, respiratory distress, persistent central cyanosis, poor tone, or bradycardia.

Assessment | Initial newborn assessment

- Term gestation, spontaneous breathing or crying, and good muscle tone.
- Heart rate, respiratory effort, central color, temperature, and responsiveness.
- Gestational age, meconium, delivery complications, multiple gestation, congenital disease, and maternal medications or substance exposure.

Assessment | Expected preductal oxygen saturation

- 2 minutes: 65-70%; 3 minutes: 70-75%; 4 minutes: 75-80%.
- 5 minutes: 80-85%; 10 minutes: 85-95%.
- Use right-hand/wrist/upper-arm preductal monitoring. These expected values apply to an infant not requiring active resuscitation.

Treatment steps

- If a non-vigorous infant does not improve after warming, drying, and stimulation within 30 seconds, move quickly to positive-pressure ventilation.
- Assess adequate ventilation primarily by visible chest movement and a rising heart rate.
- Add oxygen when the infant is not responding with an increasing heart rate or adequate breathing.
- Do not interrupt resuscitation to obtain APGAR scores; document APGAR at 1 and 5 minutes.

Medications

Epinephrine 1:10,000

0.01 mg/kg IV/IO every 3-5 minutes as needed

- For heart rate below 60 despite effective ventilation and coordinated compressions. Paramedic-only when a pulse is present; AEMT/Paramedic for confirmed pulseless arrest within approved arrest orders. Use UVC/IO access when available.

Normal saline

10 mL/kg IV/IO; may repeat once when not responding to epinephrine

- Consider suspected hypovolemia.

Warnings / precautions

- Do not routinely suction the airway; suction only when needed for obstruction.
- Do not perform routine endotracheal suctioning for meconium; provide supportive ventilation.
- Naloxone is not recommended for neonatal respiratory depression from maternal narcotics; provide ventilatory support.
- Ventilation must be effective before chest compressions are started.
- AO-02 does not specify a neonatal D10 dose. After immediate newborn stabilization, use PM-02 for the approved pediatric hypoglycemia pathway.

Clinical pearls

- A rising heart rate is the best indicator that newborn ventilation and resuscitation are effective.

- Skin-to-skin contact with the mother is the preferred warming method for a vigorous infant.
- Delay cord clamping for approximately 60 seconds unless immediate resuscitation is required.

Special populations | Premature infant

- Premature infants lose heat rapidly; use a cap, plastic wrap, thermal mattress, radiant heat, or other approved warming adjunct.
- Anticipate respiratory support, hypoglycemia, and the need for specialized neonatal destination care.

Related pathways / contacts

PM-02 Pediatric Diabetic / Hypoglycemia - After immediate newborn stabilization, open for the approved pediatric hypoglycemia and dextrose pathway.

Source and references

Claiborne County EMS approved protocol manual. | Claiborne County EMS medication formulary, linked protocol pathways, and agency release guidance.

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